Modules

```
are files containing set of codes/functions which can be included to an
application
```

Creating a Module

```
create a filename.py file and def a function there
```

Importing a Module

```
use import keyword + name of the file
    OR
from filename import function1, function2
    OR
from filename import * #import all
```

Renaming Module and Functions

```
import random as rnd
    OR
from random import randint as ri
    OR
BOTH at once
```

Import Built-In Modules

- math
- datetime
- sys
- os
- random
- statistics
- collections
- json
- re

OS Module

```
used to perform many operating system task, provide functions for creating,
changing, working directory, removing directory, fetching its content,
```

```
changing, identifying current directory
```

```
import os

os.mkdir("directory_name") # Make directory
os.chdir("path") # Change directory
os.getcwd() # Get current working directory
os.rmdir() # Remove Directory
```

Sys Module

```
provides functions and variables to manipulate different parts of python
runtime environment
```

```
import sys

sys.argv[index] # index 0 is the name of the script, index 1 is argument
passed from CMD
sys.exit() # exit sys
sys.maxsize # to know the maximum size of python data structure
sys.path # to know environment path
sys.version() # to know your python version
```

Statistic Module

```
provides functions for mathematical statistics of numeric data
```

```
import statistics

mean(list)
median(list)
mode(list)
stdev(list)
```

Math Module

```
contain many mathematical operations and constants
```

```
import math
math.pi # pi constant
math.sqrt(number) # square root
math.pow(2, 3) # 2 to the power of 3
math.floor(float) # round to the lowest
```

```
math.ceil(float) # round to the highest
math.log10(100) # log base ten of one hundred
```

String Module

```
import string

string.ascii_letters # return all ascii letters
string.digits # return digits from 0-9
string.punctuation # return all punctuations
```

Random Module

```
import random

random() # return random float from 0 to 0.999..
randint(low, high)
choice(list) # random number in the lists
```